Connecting to a Web Server

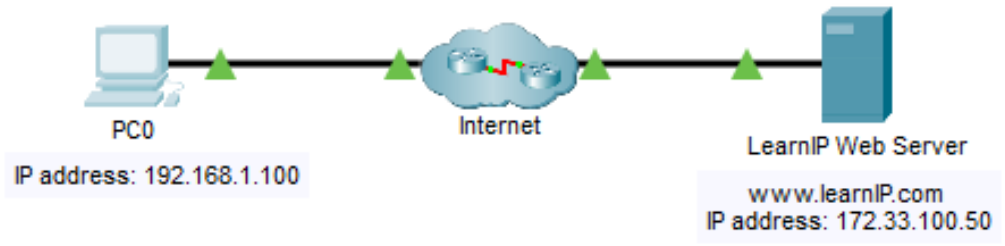
Have you ever wondered how your devices (smartphones, laptops, computers, etc.) communicate and connect over the internet? Well, I can't cover everything in this documentation, but there's certainly a lot that happens. In that unknown darkness, that vacuum you can't see, your devices perform many activities. I will only demonstrate the basic connection between devices and the internet, and how to test the connection.

Objective

The objective of this documentation is to demonstrate how IP packets are sent across the internet. I am using Cisco Packet Tracer version 9.0.1.0858. I will demonstrate using a terminal and a web browser.

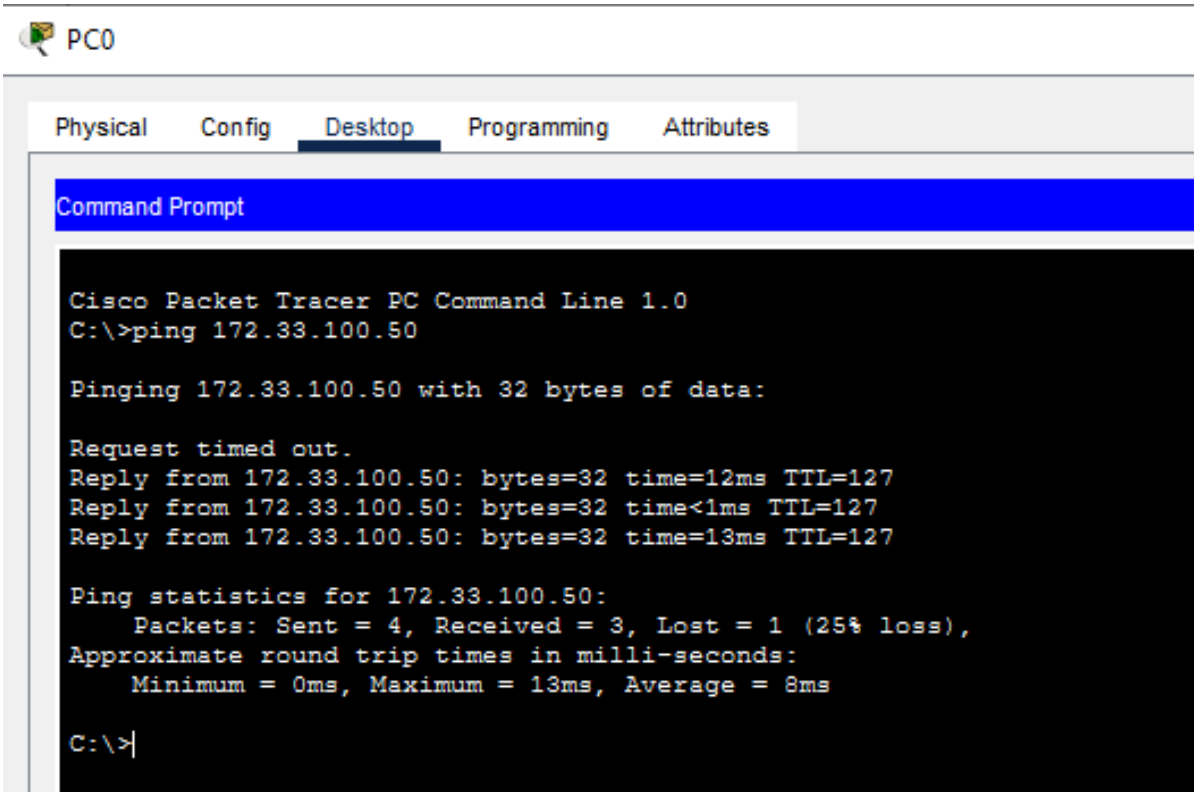
Step 1: The Setup

In this lab, I have this configuration already set up. You will find PC0 (a computer), the internet (the cloud shape), and a web server that hosts the website.



Step 2: Terminal

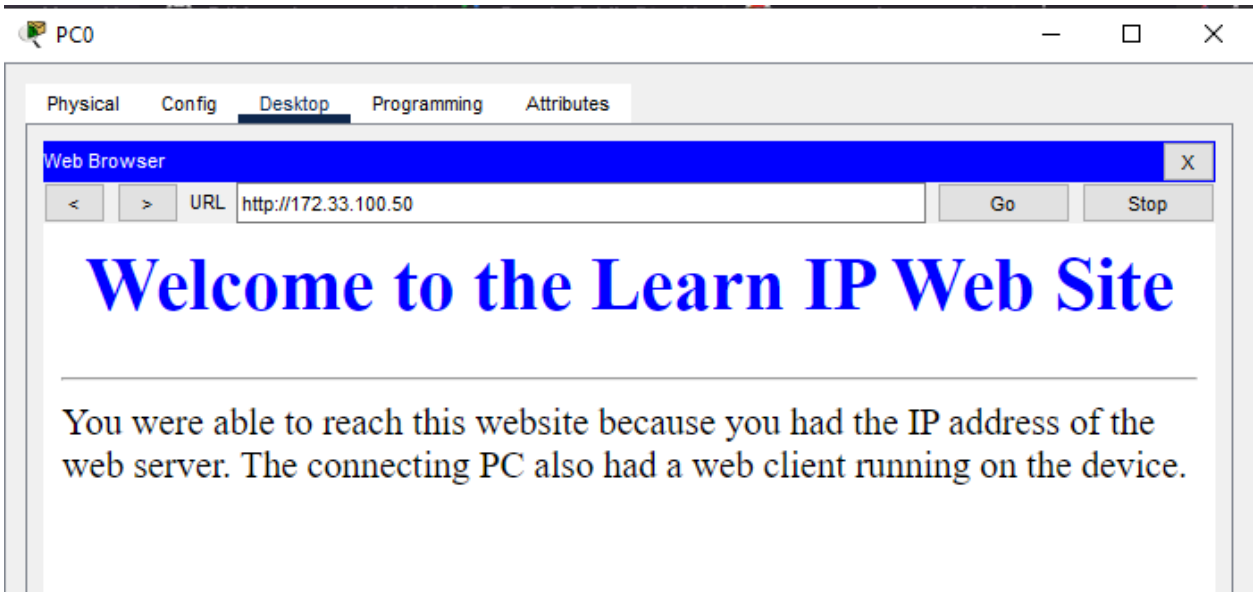
Using the terminal, I will test the connection from PC0 to the web server. Click PC0 > select the Desktop tab > Command Prompt. At the command prompt, ping the IP address of the web server by entering `ping 172.33.100.50`.



SUCCESSFUL! A reply verifies connectivity from the client to the destination web server. The reply may time out initially while the devices load and ARP is performed. Close the terminal.

Step 3: Connect to the Web Server via the Web Client

On the same Desktop tab, click Web Browser, type the same IP address you used earlier (172.33.100.50), and press Enter. The web client will connect to the web server via the IP address and open the web page.



Conclusion

This is just a simple illustration that demonstrates how devices interact using internet protocols. That is why, when you type www.google.com on your smartphone, it takes you to Google's web page.

Do a Simple Demonstration

On your phone or laptop, type 8.8.8.8 or 8.8.4.4 (Google's public IP addresses) and see what the web server shows you. You could also use a terminal to ping these addresses!